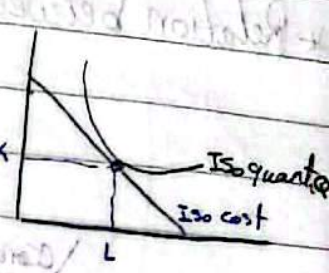
* TC: $w \cdot L + r \cdot K$
* AC: $\frac{TC}{Q \to (TP)}$
* MC: $\frac{\Delta TC}{\Delta Q} = \frac{\partial TC}{\partial Q}$
* $Q = A \cdot K^{a} \cdot L^{b}$ (cobb-douglas)

* APL usually doest ~~ee~~ zero; only at the origin point when TP, Q = zero

*Note: ex $w = 5$ $r = 10$

$TC = 5L + 10K$

slope Iso quant = slop Iscost



$MRTS = \frac{w}{r}$

$\frac{MPL}{MPK} = \frac{w}{r}$

$\Rightarrow Q = \sqrt{K \cdot L} = \sqrt{K} \times \sqrt{L}$

$Q = K^{\frac{1}{2}} \times L^{\frac{1}{2}}$

• MPL? • MPK

$MPL = \frac{\Delta Q}{\Delta L} = \frac{\partial Q}{\partial L} = \frac{1}{2} L^{-\frac{1}{2}} \cdot K^{\frac{1}{2}}$

$MPK = \frac{\Delta Q}{\Delta K} = \frac{\partial Q}{\partial K} = \frac{1}{2} K^{-\frac{1}{2}} \cdot L^{\frac{1}{2}}$

• TC ??

$TC = w \cdot L + rK$

$TC = 5L + 10K$

$MRTS = \frac{w}{r} = \frac{5}{10} = \frac{1}{2}$

$\frac{MPL}{MPK} = \frac{1}{2}$

الأس السالب بنحطه مكانه

$\frac{MPL}{MPK} = \frac{\cancel{\frac{1}{2}} L^{-\frac{1}{2}} \cdot K^{\frac{1}{2}}}{\cancel{\frac{1}{2}} K^{-\frac{1}{2}} \cdot L^{\frac{1}{2}}} = \frac{K^{\frac{1}{2}} \cdot K^{\frac{1}{2}}}{L^{\frac{1}{2}} \cdot L^{\frac{1}{2}}} = \frac{K}{L}$

$\frac{K}{L} = \frac{1}{2} \Longrightarrow L = 2K \quad , K = \frac{1}{2} L$

$\therefore Q = \sqrt{L \cdot K} = \sqrt{L \cdot \frac{1}{2} L} = \sqrt{\frac{1}{2} L^2} = \frac{1}{4} L$

$\therefore L = 4Q$

$\because K = \frac{1}{2} L$

$\therefore K = 2Q$

$\Longrightarrow TC = 5 \times 4Q + 10 \times 2Q$

$TC = 20Q + 20Q$

$= 40Q.$

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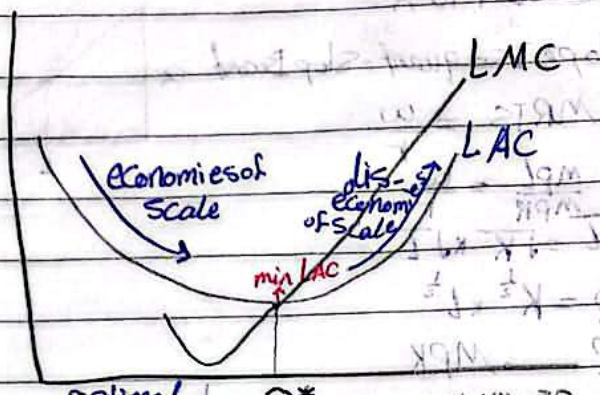
* Relation between Long run AC and Long run MC



optimal (effecient) output ← Q^* → حجم الانتاج الي ربط الشركة لاقل تكاليف

↓

MES: minimum effecient scale

economies of scal ⇒ $LAC\downarrow > MC$

diseconomies of scale ⇒ $LAC\uparrow < MC$

---

ex $TC = 1000Q - 30Q^2 + Q^3$ → تعبر عن جميع تكاليف الثابتة fixed

(1) drive LAC, LMC

$$LAC = \frac{TC}{Q} = \frac{1000Q - 30Q^2 + Q^3}{Q} = 1000 - 30Q + Q^2$$

$$LMC = \frac{\Delta TC}{\Delta Q} = \frac{\partial TC}{\partial Q} = 1000 - 60Q + 3Q^2$$

(2) determine the minimum effecient scale (MES) ⇒ LAC → min

$LAC = MC$

$1000 - 30Q + 3Q^2 = 1000 - 60Q + 3Q^2$

$-30Q + Q^2 = -60Q + 3Q^2$

$60Q - 30Q = 3Q^2 - Q^2$

$30Q = 2Q^2$

$15Q = Q^2$

$15 = Q$

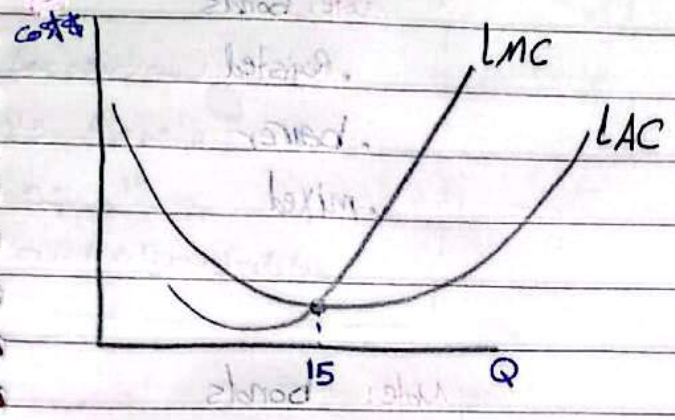
$\frac{\partial AC}{\partial Q} = zero$ لما طلب ال min بتاع حاجه نشتقها ونساوي المشتقه بصفر.

$= -30 + 2Q$

$2Q = 30$

$Q = 15$

(3) Show your answer graphically.



Note: economies of scale: >15 before 15

diseconomies of scale: <15 after 15